Nios® V/g Processor Helloworld Design

Agilex™ 5 FPGA

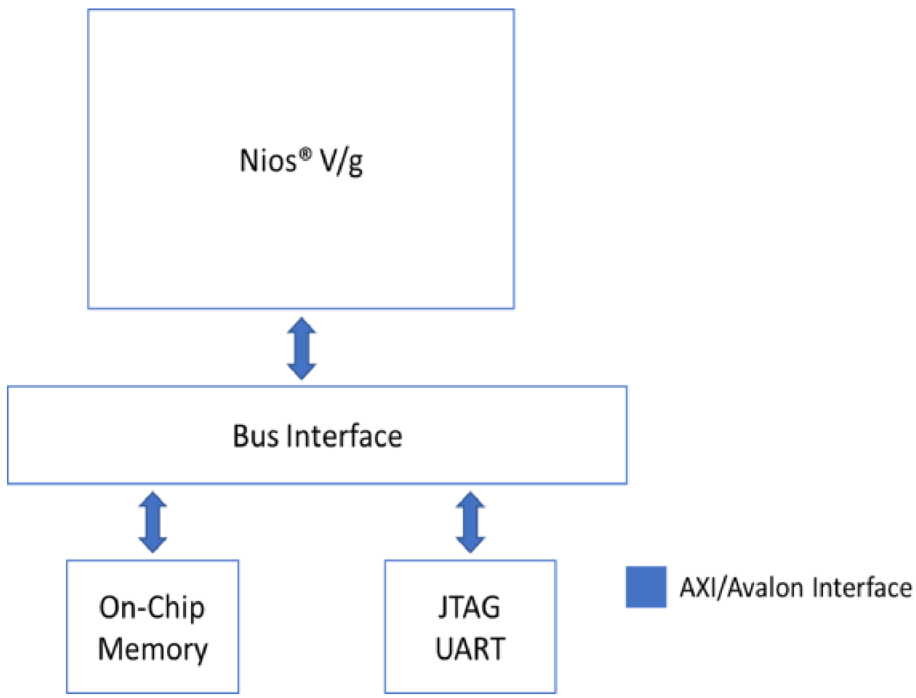
Last updated: **2024-09-30**

Contents

1.0 Theory of Operation

Nios® V/g Processor-based Helloworld example design on the Agilex™ 5 FPGA E-Series 065B Premium Development Kit (ES1) DK-A5E065BB32AES1.

Figure 1-1. Block Diagram



1.1 IP Cores

The following IPs are used in this design.

- Nios V/g soft processor core
- On Chip RAM-II
- JTAG UART
- PIO IP
- System ID

2.0 Simulating the design

Note: Please refer to the README.md file in the extracted design package from Quartus or in the sources folder of the Github repository of the design for the steps to create the design, application and generate the programing files.

- Unpackage/extract the design in your working directory.
- Simulate the design using Quartus Prime Pro tool.

3.0 Executing the design on Development Kit

3.1 Creating and validating the design

Note: Please refer to the README.md file in the extracted design package from Quartus or in the sources folder of the Github repository of the design for the steps to create the design, application and generate the programing files.

- Unpackage/extract the design in your working directory.
  Locate the "ready_to_test" folder within the package.
- The folder contains the necessary files for executing the application on the board.
  Refer to the readme file for the steps to program the application files on the board.
- Validate the design by observing the prints on the terminal.

4.0 Expected Results

The following is the output as observed on the JTAG UART terminal. The output is analogous to the logic from the application code. Users should be able to observe same output on their terminal/setup.

```
# Hello world, this is the Nios V/g cpu checking in 0...
# Hello world, this is the Nios V/g cpu checking in 1...
# Hello world, this is the Nios V/g cpu checking in 2...
# Hello world, this is the Nios V/g cpu checking in 3...
# Hello world, this is the Nios V/g cpu checking in 4...
# Hello world, this is the Nios V/g cpu checking in 5...
# Hello world, this is the Nios V/g cpu checking in 6...
# Hello world, this is the Nios V/g cpu checking in 7...
# Hello world, this is the Nios V/g cpu checking in 8...
# Hello world, this is the Nios V/g cpu checking in 9...
# Bye world!
#
```

5.0 Document Revision History

Date	Version	Changes
2024-09-30	24.3.0	• Initial release on github